

E123

Cybersecurity Fundamentals

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A regional bank's online portal, hosted in a data center built in 2019, goes offline for six hours after a flood of traffic. No customer data was viewed or changed. The bank has 85 branches. Which security goal was harmed?

- A. Confidentiality.
- B. Availability.
- C. All three, since any attack harms the whole triad.
- D. Integrity.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A lab's main security goal is to prove that test results were not altered after entry. Which control most directly supports that goal?

- A. Adding a backup generator.
- B. Encrypting the results at rest.
- C. Storing a hash of each result and comparing it to a freshly computed hash.
- D. Requiring a password to view results.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A clinic's file server is valued at \$80,000. A ransomware event would destroy 25% of its value, and the security team expects such an event once every 4 years. The server came from a vendor with a 5-star rating and has 12 TB of storage. What is the annualized loss expectancy (ALE)?

- A. \$80,000
- B. \$15,000
- C. \$20,000
- D. \$5,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

An asset's ALE is \$6,000 and its SLE is \$24,000. How often is the loss expected to occur?

- A.** 4 times a year.
- B.** Once every 6 years.
- C.** Once a year, since ALE is always an annual figure.
- D.** Once every 4 years.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A small online retailer with a popular holiday catalog buys a cyber insurance policy to cover losses if customer data is stolen. The owner also hired a new marketing lead. Which risk response does the insurance represent?

- A.** Risk transfer.
- B.** Risk elimination, because insurance removes the risk entirely.
- C.** Risk avoidance.
- D.** Risk acceptance.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A university relies on a single perimeter firewall. A student laptop infected off campus connects to the internal network and spreads malware to lab computers. The campus has 18,000 students and a new stadium. Which principle was most missing?

- A.** A stronger perimeter firewall.
- B.** Nothing, since no control can stop malware once it is inside.
- C.** Defense in depth, such as network segmentation and endpoint protection behind the firewall.
- D.** Security through obscurity.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A network technician at a warehouse that ships 5,000 orders a day needs a device that forwards traffic between two different IP subnets. The warehouse recently repainted its floor markings. At which OSI layer does that device mainly work?

- A. Layer 4, transport.
- B. Layer 3, network.
- C. Layer 2, data link.
- D. Layer 5, session.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

Two offices of a law firm with 140 attorneys need to agree on a shared secret key over the internet before encrypting large case files. The firm's IT lead prefers open-source tools. Which approach fits best?

- A. Use asymmetric encryption to exchange a symmetric key, then symmetric encryption for the files.
- B. Use asymmetric encryption for all the large files.
- C. Email the symmetric key in plain text, then encrypt the files.
- D. Hash the files before sending them.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

With public-key encryption, a sender wants only the recipient to be able to read a message. Which key does the sender use to encrypt it?

- A. The recipient's private key.
- B. The recipient's public key.
- C. The sender's private key.
- D. The sender's public key.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A hospital with a busy emergency department requires staff to enter a password and then answer a security question about their first pet. The CIO notes that login time averages 9 seconds. Does this setup provide multifactor authentication?

- A.** No, because MFA always requires a fingerprint.
- B.** Yes, because the 9-second login shows added protection.
- C.** No, because both are the same factor type: something you know.
- D.** Yes, because there are two separate prompts.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Which combination is true multifactor authentication?

- A.** Any two logins in a row, as long as the passwords differ.
- B.** A fingerprint plus a face scan.
- C.** A password plus a PIN.
- D.** A smart card plus a PIN.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

After a malware outbreak is contained at a county water utility that serves 60,000 homes, the team restores systems from clean backups and updates its recovery plan with lessons learned. The utility's board meets monthly. Which NIST CSF function does this work mainly represent?

- A.** Recover.
- B.** Govern, since the board meets monthly.
- C.** Detect.
- D.** Respond.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A regional bank's online portal, ~~hosted in a data center built in 2019~~, goes offline for six hours after a flood of traffic. No customer data was viewed or changed. ~~The bank has 85 branches.~~ Which security goal was harmed?

Target word: which CIA goal **Decides:** The service was unreachable; data was not seen or changed.

A	Adjacent category	No data was exposed.
B	Correct	Users could not reach the service when needed.
C	Absolute language	Attacks often hit one goal; the stem rules out the other two.
D	Confusion set	Integrity concerns unauthorized changes; none occurred.

2. Answer C Concept 1 · transfer

A lab's main security goal is to prove that test results were not altered after entry. Which control most directly supports that goal?

Target word: reverse: goal given, pick the control **Decides:** Proving no change = integrity.

A	Confusion set	Supports availability.
B	Adjacent category	Encryption mainly protects confidentiality.
C	Correct	Matching hashes show the data has not changed, which is integrity.
D	Inverted operation	Controls who can read data, not whether it changed.

3. Answer D Concept 2

A clinic's file server is valued at \$80,000. A ransomware event would destroy 25% of its value, and the security team expects such an event once every 4 years. The server ~~came from a vendor with a 5-star rating and has 12 TB of storage.~~ What is the annualized loss expectancy (ALE)?

Target word: ALE **Decides:** $SLE = \$80,000 \times 25\% = \$20,000$; $ARO = 1/4 = 0.25$.

A	Wrong stem number	The asset value, not the expected yearly loss.
B	Inverted operation	Uses the 75% of value that survives (\$60,000) instead of the 25% lost.
C	Dropped element	That is the SLE; it leaves out the annual rate.
D	Correct	$\$20,000 \times 0.25 = \$5,000$ per year.

4. Answer D Concept 2 · transfer

An asset's ALE is \$6,000 and its SLE is \$24,000. How often is the loss expected to occur?

Target word: solve for ARO **Decides:** $ARO = ALE / SLE = 0.25$.

A	Inverted operation	Divides SLE by ALE instead of ALE by SLE.
B	Wrong stem number	Reads the \$6,000 figure as a number of years.
C	Absolute language	ALE is annualized, but the event rate can be more or less than once a year.
D	Correct	$\$6,000 / \$24,000 = 0.25$ per year, or once every 4 years.

5. Answer A Concept 3

A small online retailer ~~with a popular holiday catalog~~ buys a cyber insurance policy to cover losses if customer data is stolen. ~~The owner also hired a new marketing lead.~~ Which risk response does the insurance represent?

Target word: risk response **Decides:** Insurance shifts the financial impact to another party.

A	Correct	Insurance moves the financial consequence to the insurer.
B	Absolute language	The breach can still happen; only the cost is shifted.
C	Confusion set	Avoidance means stopping the risky activity; the retailer still sells online.
D	Adjacent category	Acceptance means carrying the risk without further action; buying coverage is an action.

6. Answer C Concept 4

A university relies on a single perimeter firewall. A student laptop infected off campus connects to the internal network and spreads malware to lab computers. ~~The campus has 18,000 students and a new stadium.~~ Which principle was most missing?

Target word: missing principle **Decides:** Once inside, no other layer stopped the spread.

A	Dropped element	Still a single layer; the threat entered from inside.
B	Absolute language	Internal layers exist exactly to contain threats already inside.
C	Correct	Layers inside the perimeter would have limited the spread.
D	Confusion set	Hiding details is not a real layered control.

7. Answer B Concept 5

A network technician at a warehouse ~~that ships 5,000 orders a day~~ needs a device that forwards traffic between two different IP subnets. ~~The warehouse recently repainted its floor markings.~~ At which OSI layer does that device mainly work?

Target word: OSI layer of the device **Decides:** Forwarding between IP subnets is routing.

A	Adjacent category	Transport handles ports and TCP/UDP, not routing between subnets.
B	Correct	Routers use IP addresses to move traffic between networks.
C	Confusion set	Switches forward by MAC address within a network.
D	Wrong stem number	Picks up the 5 from the order count; session does not route.

8. Answer A Concept 6

Two offices of a law firm ~~with 140 attorneys~~ need to agree on a shared secret key over the internet before encrypting large case files. ~~The firm's IT lead prefers open source tools.~~ Which approach fits best?

Target word: encryption approach **Decides:** Secure key exchange plus fast bulk encryption.

A	Correct	Asymmetric solves key exchange; symmetric is efficient for bulk data.
B	Adjacent category	Works in theory but is far slower for bulk data.
C	Dropped element	Drops the need to protect the key in transit.
D	Confusion set	Hashing is one-way and checks integrity; it does not keep files secret.

9. Answer B Concept 6 · transfer

With public-key encryption, a sender wants only the recipient to be able to read a message. Which key does the sender use to encrypt it?

Target word: which key encrypts for confidentiality **Decides:** Only the recipient holds the matching private key.

A	Confusion set	The sender never has it; private keys stay with their owner.
B	Correct	Only the recipient's private key can then decrypt it.
C	Inverted operation	That creates a digital signature, which anyone with the public key can check.
D	Adjacent category	Only the sender could decrypt it, not the recipient.

10. Answer C Concept 7

A hospital ~~with a busy emergency department~~ requires staff to enter a password and then answer a security question about their first pet. ~~The CIO notes that login time averages 9 seconds.~~ Does this setup provide multifactor authentication?

Target word: is it MFA **Decides:** Both prompts are something you know.

A	Absolute language	A token or phone (something you have) also works as a second factor.
B	Camouflage	Login time does not decide factor types.
C	Correct	MFA needs different categories, such as know + have.
D	Dropped element	Drops the requirement that factors come from different categories.

11. Answer D Concept 7 · transfer

Which combination is true multifactor authentication?

Target word: reverse: pick the valid pair **Decides:** Two different factor categories.

A	Absolute language	Two passwords are still one factor type.
B	Adjacent category	Both are something you are.
C	Confusion set	Both are something you know.
D	Correct	Something you have plus something you know.

12. Answer A Concept 8

After a malware outbreak is contained at a county water utility ~~that serves 60,000 homes~~, the team restores systems from clean backups and updates its recovery plan with lessons learned. ~~The utility's board meets monthly.~~ Which NIST CSF function does this work mainly represent?

Target word: CSF function **Decides:** Containment is done; the team is restoring operations.

A	Correct	Restoring capability and improving recovery plans is Recover.
B	Camouflage	Answers the board detail; restoring systems is not governance.
C	Confusion set	Detect is about discovering events.
D	Adjacent category	Respond covers containment and analysis, which was already finished.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	4	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	The CIA triad	Confidentiality keeps data from unauthorized eyes, integrity keeps it from unauthorized change, and availability keeps it usable when needed.	Assuming every attack harms confidentiality.	Was data seen, changed, or made unreachable?
2	Quantitative risk (ALE)	$SLE = \text{asset value} \times \text{exposure factor}$; $ALE = SLE \times ARO$ (expected events per year).	Stopping at SLE, or multiplying by the number of years between events.	A dollar value, a percentage lost, and a frequency.
3	Risk responses	Avoid (stop the activity), mitigate (reduce likelihood or impact), transfer (shift the financial impact, e.g., insurance), or accept (knowingly carry it).	Thinking insurance removes the risk itself.	Insurance, outsourcing, or a decision to stop an activity.
4	Defense in depth	Use multiple independent layers so a single failure does not expose everything.	Relying on one strong perimeter.	Something got past one control and nothing else stopped it.
5	OSI layers and devices	Switches forward by MAC address at Layer 2; routers forward between IP networks at Layer 3; ports and TCP/UDP live at Layer 4.	Mixing up switches and routers.	MAC address vs. IP subnet vs. port number.
6	Symmetric, asymmetric, and hashing	Symmetric is fast for bulk data but needs a shared key; asymmetric solves key exchange; hashing is one-way and checks integrity.	Using hashing to hide data, or asymmetric for large files.	Key exchange, bulk data, or proof of no change.
7	Multifactor authentication	MFA needs factors from different categories: something you know, have, or are.	Counting two passwords or two questions as two factors.	Two prompts from the same category.
8	NIST Cybersecurity Framework functions	Govern, Identify, Protect, Detect, Respond, Recover; Recover restores capability and folds lessons into plans.	Calling restoration part of Respond.	Is the team containing an event or restoring normal operations?

Course scope (public description): Cybersecurity Fundamentals introduces learners to the core principles and modern significance of cybersecurity in organizational and technological contexts. Learners apply foundational networking concepts and security methods to develop strategies for securing systems and conduct risk assessments using established industry frameworks, building the analytical foundation needed for advanced study and professional practice in cybersecurity.